
COLLEGE MANAGEMENT SYSTEM FOR PUNJAB GROUP OF COLLEGES (PGC)

Course: Database Systems

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PROJECT OVERVIEW

Project Title:

College Management System for Punjab Group of Colleges (PGC).

Brief Description:

A computerized system designed to manage academic and administrative operations like student records, fees, and results.

Problem Statement:

Current manual processes are slow, involve too much paperwork, and are prone to human errors.

Target Users:

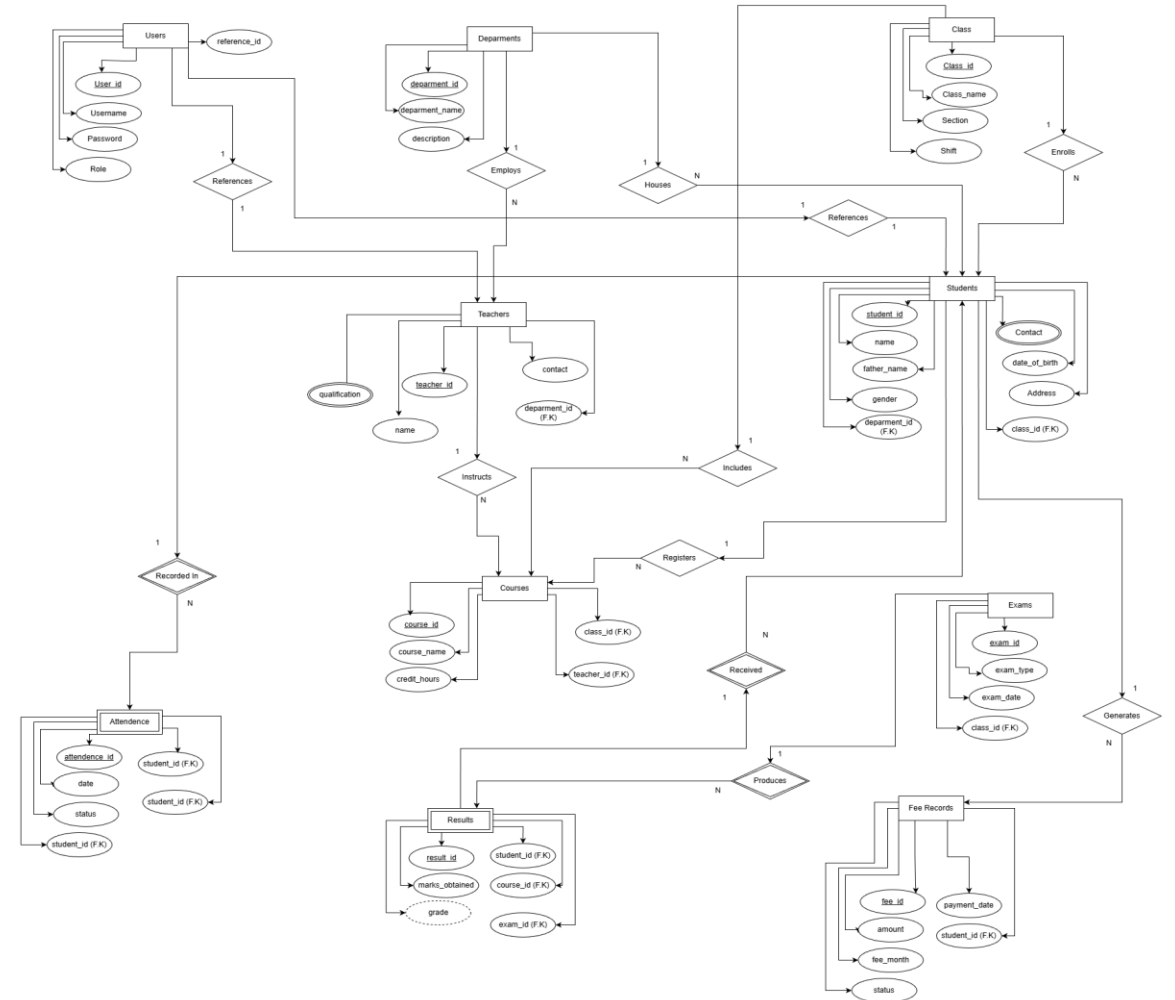
Admin staff, teachers, students, the accounts department, and the examination department.

SYSTEM FUNCTIONALITIES

- 1. Add and Update Profiles:**
Manage detailed records for both students and teachers.
- 2. Section Assignment:**
Assign students to specific sections and classes.
- 3. Course Management:**
Create and manage lists of courses offered.
- 4. Attendance Tracking:**
Allow teachers to mark and view daily student attendance.
- 5. Fee Management:**
Record fee payments, generate vouchers, and track pending dues.
- 6. Exam Scheduling:**
Create and manage schedules for midterm and final exams.
- 7. Result Generation:**
Enter student marks and automatically generate result cards.
- 8. Department Management:**
Add and manage different college departments.
- 9. Role-Based Login:**
Secure login system for Admins, Teachers, and Students.
- 10. Reporting:**
Generate automated reports for attendance, fees, and performance.

DATABASE DESIGN

Table Name	Primary Key (PK)	Attributes & Foreign Keys (FK)
Departments	<u>department_id</u>	department_name, description
Classes	<u>class_id</u>	class_name, section, shift
Students	<u>student_id</u>	name, father_name, gender, date_of_birth, contact, address, department_id (FK) , class_id (FK)
Teachers	<u>teacher_id</u>	name, qualification, contact, department_id (FK)
Courses	<u>course_id</u>	course_name, credit_hours, class_id (FK) , teacher_id (FK)
Attendance	<u>attendance_id</u>	date, status, student_id (FK) , course_id (FK)
Exams	<u>exam_id</u>	exam_type, exam_date, class_id (FK)
Results	<u>result_id</u>	marks_obtained, grade, student_id (FK) , course_id (FK) , exam_id (FK)
Fee Records	<u>fee_id</u>	amount, fee_month, status, payment_date, student_id (FK)
Users	<u>user_id</u>	username, password, role, reference_id



RELATIONSHIPS & CONSTRAINTS

■ Relationships (1:N - One-to-Many)

Department to Students/Teachers: One department can have multiple students and multiple teachers, but a student or teacher belongs to one specific department.

Class to Students: One class (e.g., ICS Section A) can have many students enrolled in it.

Class to Subjects: One class can be taught multiple subjects (courses).

Teacher to Subjects: One teacher can be assigned to teach multiple different subjects.

Student to Records: A single student will have multiple records for Attendance, Fee Payments, and Results over time.

Subject to Students: A specific subject is taught to a group of multiple students.

Exam to Results: One exam (like a Midterm) generates many individual result records for different students.

■ Key Constraints Used

Primary Keys (PK): Used in every table to ensure each record (like a specific student_id) is unique and never duplicated.

Foreign Keys (FK): These are the "bridges" that create the relationships mentioned above by linking IDs across tables.

NOT NULL: Applied to critical fields like name and username to ensure they are never left blank during data entry.

UNIQUE: Applied to username in the Users table so that no two people can have the same login name.

IMPLEMENTATION & CONCLUSION

Types of SQL Queries Implemented

Data Definition (DDL): Used CREATE DATABASE and CREATE TABLE commands to establish the schema.

Integrity Constraints: Implemented PRIMARY KEY for uniqueness, FOREIGN KEY for relational integrity, and NOT NULL or UNIQUE for data validation.

Data Manipulation (DML): Inserted 10 sample records into each table using INSERT INTO to simulate a real college environment.

Data Retrieval (DQL):

Joins: Combined multiple tables (e.g., Students, Departments, and Classes) to generate comprehensive student profiles.

Filtering & Sorting: Used WHERE clauses for specific searches (like finding PhD teachers) and ORDER BY for organized lists.

Aggregations: Utilized COUNT, SUM, and AVG to generate financial and administrative reports.

Learning Outcomes

Gained hands-on experience in translating a business proposal into a functional Relational Database Management System (RDBMS).

Mastered the process of **Normalization** to eliminate data redundancy and ensure data consistency across 10 different entities.

Developed proficiency in writing complex SQL joins and aggregate functions to extract meaningful information from raw data.

Future Improvements

Automation: Implementing SQL Triggers to automatically generate fee vouchers when a student is admitted.

Security: Adding advanced encryption for user passwords within the users table.

Front-end Integration: Developing a web-based dashboard for easier data entry and report viewing.

Conclusion:

This project successfully transitions the Punjab Group of Colleges from manual operations to a digital College Management System. By defining 10 core tables, including Students, Teachers, and Results, we established a design that manages everything from daily attendance to fee tracking. We applied 3NF normalization to ensure data accuracy and utilized SQL constraints like Primary and Foreign Keys to link entities such as Departments and Classes.

Ultimately, the implementation of complex SQL joins and aggregate queries proves that the system is a reliable and efficient tool for modern academic and financial administration.



THANK YOU